Month	Monthly milk production: pounds per cow.
1962-01	589
1962-02	561
1962-03	640
1962-04	656
1962-05	727
1962-06	697
1962-07	640
1962-08	599
1962-09	568
1962-10	577
1962-11	553
1962-12	582
1963-01	600
1963-02	566
1963-03	653
1963-04	673
1963-05	742
1963-06	716
1963-07	660
1963-08	617
1963-09	583
1963-10	587
1963-11	565
1963-12	598
1964-01	628
1964-02	618
1964-03	688
1964-04	705
1964-05	770
1964-06	736
1964-07	678
1964-08	639
1964-09	604
1964-10	611
1964-11	594
1964-12	634
1965-01	658
1965-02	622
1965-03	709
1965-04	722
1965-05	782
1965-06	756
1965-07	702
1965-08	653
1965-09	615
1965-10	621
1965-11	602
1965-12	635
1966-01	677
1966-02	635
1966-03	736
1966-04	755
1966-05	811
1966-06	798
1966-07	735
1966-08	697
1966-09	661
1966-10	667
1966-11	645
1966-12	688
1967-01	713
1967-02	667
1967-03	762
1967-04	784
1967-05	837
1967-06	817
1967-07	767
1967-08	722
1967-09	681
1967-10	687
1967-11	660
1967-12	698
1968-01	717

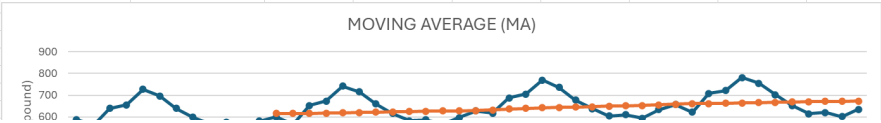
Year	Month	Milk production	t	MA	CMA	SN + IR	SNt	Dt	TRt	Y-hat = TRt+SNt	Error	Error^2
1962	1	589	1				-0.0256235738	589.0256236	615.816638	615.7910144	-26.79101443	717.758454
	2	589	2				-0.07019703063	589.070197	617.032776	616.962579	-27.96257897	781.9058226
	3	640	3				0.05439019773	639.9456098	618.248914	618.3033042	21.6966958	470.7466087
	4	656	4				0.07740177532	655.9225982	619.465052	619.5424538	36.45754622	1329.152677
	5	727	5				0.1753621244	726.8246379	620.68119	620.8565521	106.1434479	11266.43153
	6	697	6				0.1279300953	696.8720699	621.897328	622.0252581	74.9747419	5621.211924
	7	640	7				0.03820057999	639.9617994	623.113466	623.1516666	16.84833342	283.866339
	8	599	8				-0.03177736514	599.0317774	624.329604	624.2978266	-25.29782663	639.9800324
	9	568	9				-0.0888194289	568.0888194	625.545742	625.4569226	-57.45692257	3301.297951
	10	577	10				-0.08194314933	577.0819431	626.76188	626.6799369	-49.67993685	2468.096125
	11	553	11				-0.1135421185	553.1135421	627.978018	627.8644759	-74.86447588	5604.689749
	12	582	12				-0.06138210647	582.0613821	629.194156	629.1327739	-47.13277389	2221.498375
1963	1	600	13	618.0833333	618.5416667	-18.54166667	-0.0256235738	600.0256236	630.410294	630.3846704	-30.38467043	923.2281969
	2	566	14	619	618.0416667	-52.04166667	-0.07019703063	566.070197	631.626432	631.556235	-65.55623497	4297.619943
	3	653	15	617.0833333	617.625	35.375	0.05439019773	652.9456098	632.84257	632.8969602	20.1030398	404.1322093
	4	673	16	618.1666667	618.875	54.125	0.07740177532	672.9225982	634.058708	634.1361098	38.86389022	1510.401963
	5	742	17	619.5833333	620.2083333	121.7916667	0.1753621244	741.8246379	635.274846	635.4502081	106.5497919	11352.85815
	6	716	18	620.8333333	621.625	94.375	0.1279300953	715.8720699	636.490984	636.6189141	79.3810859	6301.356799
	7	660	19	622.4166667	623.25	36.75	0.03820057999	659.9617994	637.707122	637.7453226	22.25467742	495.2706671
	8	617	20	624.0833333	624.8333333	-7.833333333	-0.03177736514	617.0317774	638.92326	638.8914826	-21.89148263	479.237012
	9	583	21	625.5833333	626.2083333	-43.20833333	-0.0888194289	583.0888194	640.139398	640.0505786	-57.05057857	3254.768515
	10	587	22	626.8333333	627.25	-40.25	-0.08194314933	587.0819431	641.355536	641.2735929	-54.27359285	2945.622881
	11	565	23	627.6666667	628.1666667	-63.16666667	-0.1135421185	565.1135421	642.571674	642.4581319	-77.45813188	5999.762195
	12	598	24	628.6666667	629.3333333	-31.33333333	-0.06138210647	598.0613821	643.787812	643.7264299	-45.72642989	2090.906391
1964	1	628	25	630	631.1666667	-3.166666667	-0.0256235738	628.0256236	645.00395	644.9783264	-16.97832643	288.2635682
	2	618	26	632.3333333	634.5	-16.5	-0.07019703063	618.070197	646.220088	646.149891	-28.14989097	792.4163616
	3	688	27	636.6666667	638.125	49.875	0.05439019773	687.9456098	647.436226	647.4906162	40.5093838	1641.010176
	4	705	28	639.5833333	640.9166667	64.08333333	0.07740177532	704.9225982	648.652364	648.7297658	56.27023422	3166.33926
	5	770	29	642.25	643.4166667	126.5833333	0.1753621244	769.8246379	649.868502	650.0438641	119.9561359	14389.47453
	6	736	30	644.5833333	645.4166667	90.58333333	0.1279300953	735.8720699	651.08464	651.2125701	84.7874299	7188.90827
	7	678	31	646.25	647	31	0.03820057999	677.9617994	652.300778	652.3389786	25.66102142	658.4880203
	8	639	32	647.75	648.6666667	-9.666666667	-0.03177736514	639.0317774	653.516916	653.4851386	-14.48513863	209.8192413
	9	604	33	649.5833333	650.4583333	-46.45833333	-0.0888194289	604.0888194	654.733054	654.6442346	-50.64423457	2564.838495
	10	611	34	651.3333333	652.3333333	-41.33333333	-0.08194314933	611.0819431	655.949192	655.8672489	-44.86724885	2013.070019
	11	594	35	653.3333333	654.5416667	-60.54166667	-0.1135421185	594.1135421	657.16533	657.0517879	-63.05178788	3975.527955
	12	634	36	655.75	657.25	-23.25	-0.06138210647	634.0613821	658.381468	658.3200859	-24.32008589	591.4665779
1965	1	658	37	658.75	660	-2	-0.0256235738	658.0256236	659.597606	659.5719824	-1.571982426	2.471128748
	2	622	38	661.25	661.4166667	-39.41666667	-0.07019703063	622.070197	660.813744	660.743547	-38.74354697	1501.062432
	3	709	39	661.5833333	662.4583333	46.54166667	0.05439019773	708.9456098	662.029882	662.0842722	46.9157278	2201.085515
	4	722	40	663.3333333	664.0416667	57.95833333	0.07740177532	721.9225982	663.24602	663.3234218	58.67657822	3442.940832
	5	782	41	664.75	665.25	116.75	0.1753621244	781.8246379	664.462158	664.6375201	117.3624799	13773.95168
	6	756	42	665.75	666.5833333	89.41666667	0.1279300953	755.8720699	665.678296	665.8062261	90.1937739	8134.916851
	7	702	43	667.4166667	668.4166667	33.58333333	0.03820057999	701.9617994	666.894434	666.9326346	35.06736542	1229.720118
	8	653	44	669.4166667	670	-17	-0.03177736514	653.0317774	668.110572	668.0787946	-15.07879463	227.3700476
	9	615	45	670.5833333	671.0416667	-56.04166667	-0.0888194289	615.0888194	669.32671	669.2378906	-54.23789057	2941.748774
	10	621	46	671.5	671.9166667	-50.91666667	-0.08194314933	621.0819431	670.542848	670.4609049	-49.46090485	2446.381109
	11	602	47	672.3333333	672.6666667	-70.66666667	-0.1135421185	602.1135421	671.758986	671.6454439	-69.64544388	4850.487853
	12	635	48	673	673	-38	-0.06138210647	635.0613821	672.975124	672.9137419	-37.91374189	1437.451824
											Total	154431.0112

Unadjusted Seasonal				
Month	1962	1963	1964	SNt
1	-16.20833333	-3.166666667	-2	-7.125
2	-50.875	-16.5	-39.41666667	-35.59722222
3	35.375	49.875	46.54166667	43.93055556
4	54.125	64.08333333	57.95833333	58.72222222
5	121.7916667	126.5833333	116.75	121.7083333
6	94.375	90.58333333	89.41666667	91.45833333
7	36.75	31	33.58333333	33.77777778
8	-7.833333333	-9.666666667	-17	-11.5
9	-43.20833333	-46.45833333	-56.04166667	-48.56944444
10	-40.25	-41.33333333	-50.91666667	-44.16666667
11	-63.16666667	-60.54166667	-70.66666667	-64.79166667
12	-31.33333333	-23.25	-38	-30.86111111

Month	Unadjusted Seasonal	Adjusted Seasonal
1	-7.902777778	-16.72106481
2	-35.98611111	-44.80439815
3	43.93055556	35.11226852
4	58.72222222	49.90393519
5	121.7083333	112.8900463
6	91.45833333	82.6400463
7	33.77777778	24.95949074
8	-11.5	-20.31828704
9	-48.56944444	-57.38773148
10	-44.16666667	-52.9849537
11	-64.79166667	-73.6099537
12	-30.86111111	-39.67939815
Total	105.8194444	0
Mean	8.818287037	0

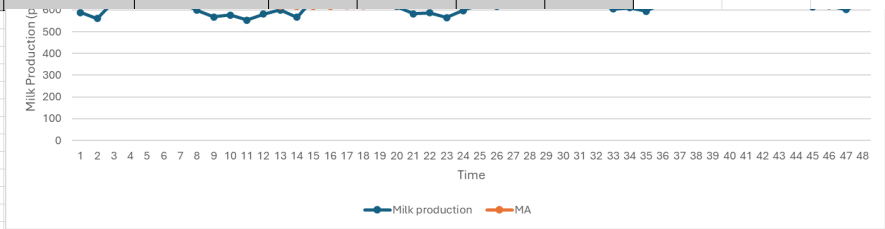
	Coefficients
Intercept (β0)	614.6005
X Variable 1 (β1)	1.216138
MSE	3217.312732

t	Milk production	Y-hat = TRt+SNt	clt+irt	clt	irt
1	589	615.7910144	-26.79101443		
2	561	616.962579	-55.96257897		
3	640	618.3033042	21.6966958	-20.3522992	42.048995
4	656	619.5424538	36.45754622	0.7305543525	35.72699187
5	727	620.8565521	106.1434479	54.76589663	51.37755124

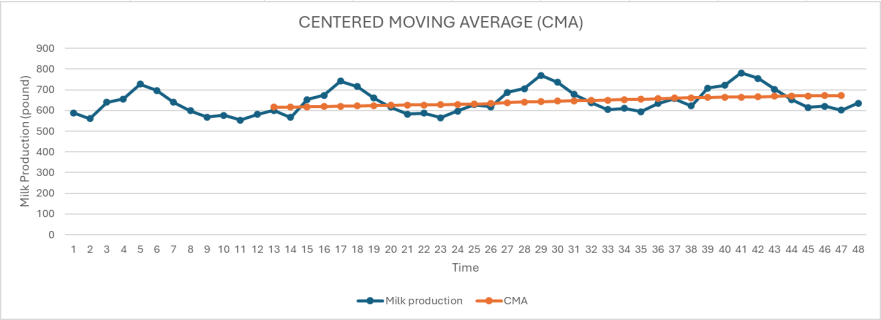


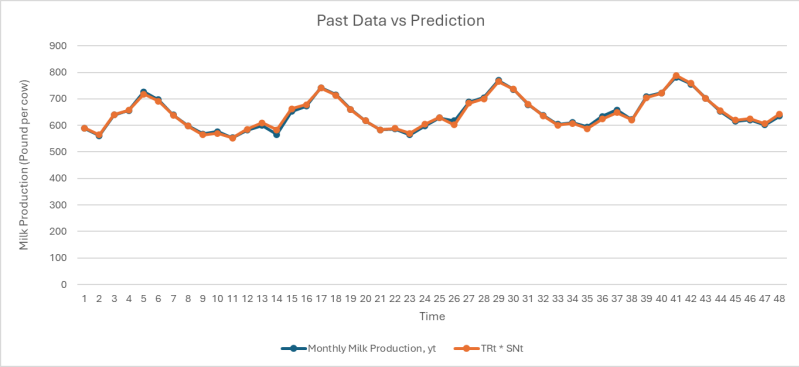
Month	Monthly milk production: pounds per cow.
1968-02	696
1968-03	775
1968-04	796
1968-05	858
1968-06	826
1968-07	783
1968-08	740
1968-09	701
1968-10	706
1968-11	677
1968-12	711
1969-01	734
1969-02	690
1969-03	785
1969-04	805
1969-05	871
1969-06	845
1969-07	801
1969-08	764
1969-09	725
1969-10	723
1969-11	690
1969-12	734
1970-01	750
1970-02	707
1970-03	807
1970-04	824
1970-05	886
1970-06	859
1970-07	819
1970-08	783
1970-09	740
1970-10	747
1970-11	711
1970-12	751
1971-01	804
1971-02	756
1971-03	860
1971-04	878
1971-05	942
1971-06	913
1971-07	869
1971-08	834
1971-09	790
1971-10	800
1971-11	763
1971-12	800
1972-01	826
1972-02	799
1972-03	890
1972-04	900
1972-05	961
1972-06	935
1972-07	894
1972-08	855
1972-09	809
1972-10	810
1972-11	766
1972-12	805
1973-01	821
1973-02	773
1973-03	883
1973-04	898
1973-05	957
1973-06	924
1973-07	881
1973-08	837
1973-09	784
1973-10	791
1973-11	760
1973-12	802
1974-01	828
1974-02	778

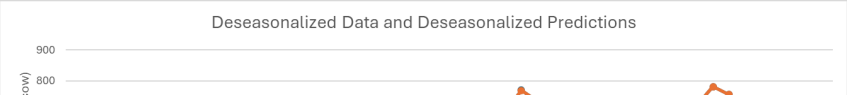
Year	Month	Milk production	t	MA	CMA	SN + IR	SNt	Dt	TRt	Y-hat = TRt+SNt	Error	Error^2
6	697	622.0252581	74.9747419	72.52524533	2.44949657							
7	640	623.1516666	16.84833342	65.98884107	-49.14050765							
8	599	624.2978266	-25.29782663	22.1750829	-47.47290953							
9	568	625.4569226	-57.45692257	-21.96880526	-35.48811731							
10	577	626.6799369	-49.67993685	-44.14489535	-5.535041498							
11	553	627.8644759	-74.86447588	-60.66711177	-14.19736411							
12	582	629.1327739	-47.13277389	-57.22572888	10.09295498							
13	600	630.3846704	-30.38467043	-50.7939734	20.40930297							
14	566	631.556235	-65.55623497	-47.69122643	-17.86500854							
15	653	632.8969602	20.1030398	-25.27928853	45.38232833							
16	673	634.1361098	38.86389022	-2.196434981	41.06032521							
17	742	635.4502081	106.5497919	55.17224063	51.37755124							
18	716	636.6189141	79.3810859	74.93158933	4.44949657							
19	660	637.7453226	22.25467742	69.39518507	-47.14050765							
20	617	638.8914826	-21.89148263	26.5814269	-48.47290953							
21	583	640.0505786	-57.05057857	-18.8957946	-38.15478398							
22	587	641.2735929	-54.27359285	-44.40521802	-9.868374832							
23	565	642.4581319	-77.45813188	-62.92743443	-14.53069745							
24	598	643.7264299	-45.72642989	-59.15271821	13.42628832							
25	628	644.9783264	-16.97832643	-46.72096273	29.74263631							
26	618	646.149891	-28.14989097	-30.28488243	2.13499146							
27	688	647.4906162	40.5093838	-1.539611198	42.048995							
28	705	648.7297658	56.27023422	22.87657569	33.39365854							
29	770	650.0438641	119.9561359	72.2452513	47.71088457							
30	736	651.2125701	84.7874299	87.0046	-2.217170097							
31	678	652.3389786	25.66102142	76.80152907	-51.14050765							
32	639	653.4851386	-14.48513863	31.9877709	-46.47290953							
33	604	654.6442346	-50.64423457	-13.15611726	-37.48811731							
34	611	655.8672489	-44.86724885	-36.66554069	-8.201708165							
35	594	657.0517879	-63.05178788	-52.85442377	-10.19736411							
36	634	658.3200859	-24.32008589	-44.07970754	19.75962165							
37	658	659.5719824	-1.571982426	-29.64795207	28.07596964							
38	622	660.743547	-38.74354697	-21.5452051	-17.19834187							
39	709	662.0842722	46.9157278	2.200066136	44.71566167							
40	722	663.3234218	58.67657822	22.28291969	36.39365854							
41	782	664.6375201	117.3624799	74.31826197	43.04421791							
42	756	665.8062261	90.1937739	88.74427733	1.44949657							
43	702	666.9326346	35.06736542	80.87453973	-45.80717431							
44	653	668.0787946	-15.07879463	36.72744823	-51.80624286							
45	615	669.2378906	-54.23789057	-11.41643993	-42.82145064							
46	621	670.4609049	-49.46090485	-39.59253002	-9.868374832							
47	602	671.6454439	-69.64544388	-57.7814131	-11.86403078							
48	635	672.9137419	-37.91374189	-52.34003021	14.42628832							

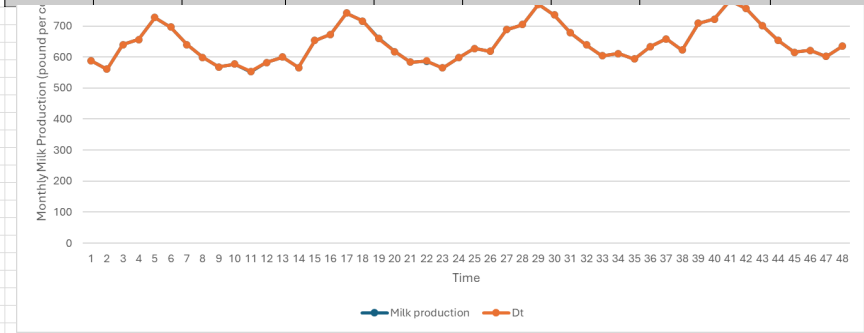


The forecasting model is adaptable, as it closely follows the actual data's trends.







Month	Monthly milk production: pounds per cow.
1974-03	889
1974-04	902
1974-05	969
1974-06	947
1974-07	908
1974-08	867
1974-09	815
1974-10	812
1974-11	773
1974-12	813
1975-01	834
1975-02	782
1975-03	892
1975-04	903
1975-05	966
1975-06	937
1975-07	896
1975-08	858
1975-09	817
1975-10	827
1975-11	797
1975-12	843

Year	Month	Milk production	t	MA	CMA	SN + IR	SNt	Dt	TRt	Y-hat = TRt+SNt	Error	Error^2